[INCH-POUND]
A-A-59229
May 20, 1998
SUPERSEDING
(see 6.6)

COMMERCIAL ITEM DESCRIPTION

CAP, VENT, FILLER, BATTERY

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This CID covers requirements for vent/filler cap for automotive type batteries. The vent/filler cap is intended to be fully functionable for waterproof, low-maintenance, lead-acid batteries and perform either in wet or dry charged conditions.

2. SALIENT CHARACTERISTICS

2.1 Materials. Materials shall be in accordance with the manufacturer's specifications for the vent/filler cap and shall meet all of the requirements specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see 3.1, 6.3, and 6.4).

2.2 Design and construction. The vent/filler cap shall conform to the manufacturer's design and construction except for the nominal dimensions and tolerances which shall conform to figure 1. The check valve shall not leak more than 2 drops of water in 10 seconds when under 78.7 inches [2.0 meters (m)] head of water and shall be designed to open before internal pressures reach 1.5 pounds per square inch (psi) [10.3 kilopascals (kPa)] in excess of the internal pressure. When tested as specified, the vent/filler cap shall withstand temperatures from -34 degrees Fahrenheit (°F) through 250°F [-48 degrees Celsius (°C) through 121°C] without cracking, melting, or other damage. To determine conformance to these salient characteristics, each vent/filler cap under test shall be visually inspected initially to the nominal dimensions and tolerances of figure 1 and then exposed to an ambient temperature of -34°F ± 2°F [-48°C ± 1°C] for 2 hours. Following the cold exposure, the vent/filler cap shall be removed from the cold source and immediately exposed to an ambient temperature of 250°F ± 5°F [121°C ± 3°C] for 90 minutes. Immediately upon removal from the heat source, each cap shall be inspected for evidence of damage such as cracking,

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melting, or other deformities. After a cooling period of one hour, each vent/filler cap shall be inserted in a fixture or dummy battery arranged so that water pressure can be applied on the outside of the fixture and air pressure on the lower side. Using this fixture, the upper side of each vent/filler cap shall be subjected to a water pressure which exceed the air pressure on the lower side by the equivalent of a water head of 78.7 inches [2.0 meters (m)] (if air pressure is atmospheric). The number of drops of water that leak through each vent/filler cap and the time required (not to be less than one minute) shall be recorded. The water pressure shall then be released and the air pressure on the lower side increased to .5 psi [3.45 kPa] above the air pressure existing on the upper side. Each vent/filler cap shall open under the specified pressure.

2.3 Absorbent resistant. The vent/filler cap shall be nonabsorbent and acid-resistant.

2.4 Fire retardation. Vent/filler cap shall not permit ignition of flammable gases in the test battery when tested in accordance with SAE J1495.

2.5 Electrolyte retention. The vent/filler cap shall not allow electrolyte to leak from the batteries filled to the specified level with sulfuric battery acid (electrolyte) or equivalent and tilted 45° in the major axis (X) and minor axis (Y) for 30 seconds per axis. The vent/filler cap shall not allow leakage or spillage from the batteries when rotated on those axes.

2.6 Extreme temperature resistance. The vent/filler cap shall evidence no cracking or other damage due to temperature change between 190°F through -65°F [88°C through -54°C], and no more than .1 psi [0.69 kPa] pressure drop in 30 seconds, with .5 psi [3.45 kPa] pressure applied to each cell individually after exposure to extreme test temperature. To verify this characteristic the vent caps installed to a dummy battery or fixture shall be subjected to two thermal shocks with an internal pressure of .5 psi [3.45 kPa] applied to each cell for 30 seconds before and after each cycle. Drop in pressure for the individual cells shall be regarded as an indication of leakage. The following procedure for one cycle shall be used:

a. Batteries shall be placed in an ambient temperature of -65°F $\pm$ 2°F [-54°C $\pm$ 1°C] for 24 hours, or until electrolyte is stabilized.
b. The batteries shall then be placed in an ambient air temperature of 190°F $\pm$ 2°F [88°C $\pm$ 1°C] for 24 hours.
c. The batteries shall be allowed to cool gradually to 81°F $\pm$ 11°F [27°C $\pm$ 6°C] for 24 hours.

2.7 Workmanship. Vent/filler cap shall be processed in such a matter as to be uniform in quality and free of defects that will affect their life, serviceability, or appearance. Vent/filler cap shall be free of cracks, leaks, and broken vent/filler caps.

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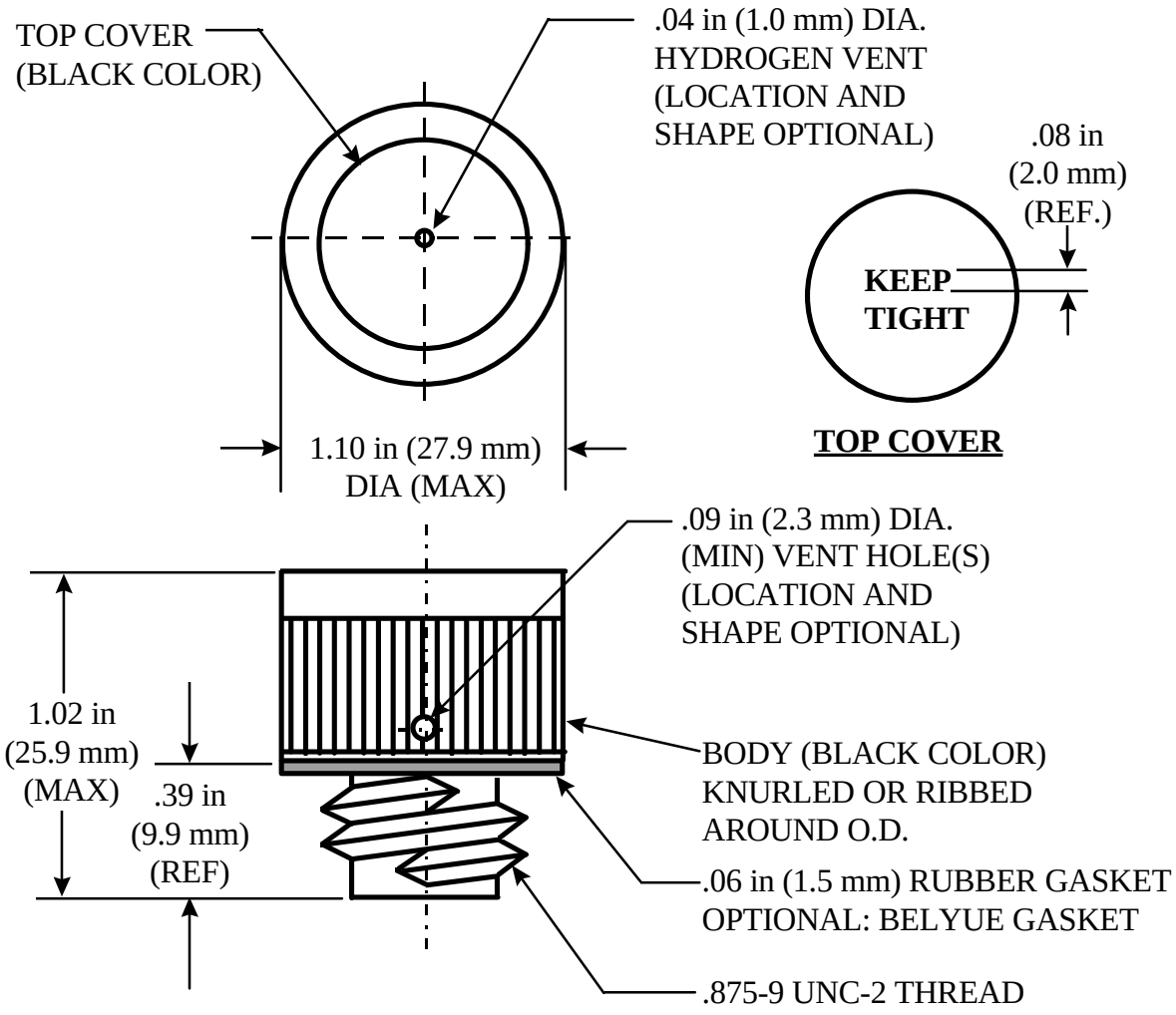

NOTES:

1. Unless otherwise specified, dimensions are in inches.
2. Tolerances are ± .015 in. (0.38 mm).
3. Unless otherwise specified, material shall be ultraviolet resistant.
4. Letters on top cover shall be bold gothic, .02 in. (0.5 mm) maximum relief.

FIGURE 1: Vent/filler plug cap.

2.8 Identification and markings. Identification and markings of the vent/filler cap shall be permanent and legible and shall include, as a minimum, the contractors identification code (CAGE) and contractors commercial part number.

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3. REGULATORY REQUIREMENTS

3.1 Recovered material. The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5. PACKAGING. Preservation, packaging, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Addresses for obtaining copies of reference documents.

6.1.1 Government publications. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 Non-Government documents. Copies of SAE J1495 "Test Procedure for Battery Flame Retardant Venting System" (DoD adopted) are available from the Society of Automotive Engineers (SAE), Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation.
c. Quantity of caps required.
d. Packaging requirements.

6.3 Gasket material(s). Previous gaskets were manufactured of rubber or belyue gasket.

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6.4 Vent/filler cap material. Previous vent/filler caps were manufactured of polypropylene because of its acid-resistant and ultraviolet resistant properties.

6.5 Electrolyte (battery acid). Previous batteries (6TL or 6TLFP), with vent/filler caps, were filled with O-S-801E, Sulfuric Acid, Electrolyte (for Storage Batteries); Class 3. This electrolyte is formulated of sulfuric acid with the specific gravity of 1.280 at 60.1°F/60.1°F [15.6°C/15.6°C]. The electrolyte should not exceed the impurity level specified in table I.

TABLE I. Maximum impurities.

Impurity	Percent by weight Class 3
Organic matter	*
Platinum	*
Fixed residue	0.013
Sulfurous acid	0.0016
Iron	0.0020
Copper	0.0020
Zinc	0.0016
Arsenic	0.00004
Antimony	0.00004
Selenium	0.0008
Nickel	0.00004
Manganese	0.000008
Nitrate	0.00020
Ammonium	0.0004
Chloride	0.0004

6.6 Supersession data. This CID supersedes vent/filler cap MS52149-3 of MS52149G(AT), dated 8 September 1994, and includes the requirements for vent/filler caps from MIL-B-62346D(AT), dated 8 September 1994. These documents have not been cancelled but have been inactivated for new design.

MILITARY INTERESTS:

Custodian:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Preparing Activity:
Army - AT

(Project 6140-0880)

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